

Who is moving to Ireland, and what housing outcomes do they face?

I started with a fairly simple question, if we look at recent immigration to Ireland by citizenship, do the groups who are arriving look similar once housing is brought into the picture, or do they start to separate quite quickly? I did not want to make a single static chart because that would only answer one part of the question. A count chart alone would show which groups are largest, but not whether those groups are younger, more likely to rent, or facing higher rents. For that reason, I built a small interactive dashboard instead of a single visual.

The dashboard has four linked views. The first shows the number of recent immigrants by citizenship, the second shows age composition within each citizenship group, the third shows housing tenure composition, and the fourth shows average weekly rent with a national reference line. A dropdown lets the viewer focus on one citizenship group across the whole dashboard. That interaction is deliberately simple. I considered doing something more elaborate, but for this dataset a single control felt cleaner and easier to understand.

The recent immigration count data comes from the CSO's F5020 table on people who were living outside the State one year earlier, while the wider interpretation of immigration and citizenship is drawn from the Census 2022 Profile 5 releases on immigration and citizenship (CSO, 2023a, 2023b).

The data is organised into separate processed CSV files rather than one raw file. I cleaned category labels so the same citizenship names could be reused across panels, and I calculated percentages within each citizenship group for the age and housing charts. That made it possible to use proportional stacked bars instead of raw counts, which was more useful for comparison. (Financial Times, no date). The result is still based on a flat table structure with a mix of nominal and quantitative attributes, which suits this kind of dashboard well (Munzner, 2014). The main categorical fields are citizenship, age group, occupancy type, and landlord type, while the main quantitative fields are counts, within-group percentages, and average weekly rent. That combination seemed like an ideal fit for multiple views because instead of just ranking categories, I wanted to compare how their internal makeup changes across measures (Matthews, 2026e, 2026c). Citizenship here means the census citizenship category rather than birthplace, so the "Ireland" group should be read as Irish citizenship rather than automatically as Ireland born returnees (CSO, 2023a, 2023b).

For the first panel I used a sorted horizontal bar chart. That choice was quite straightforward. Bars are my preferred option when the task is to compare magnitudes across categories and sorting them immediately makes the ranking visible (Cleveland and McGill, 1984; Few, 2012). Horizontal bars also work better than vertical ones here because some citizenship labels are long. Rotating labels would have made the chart harder to read for no real benefit. The point of this chart is not subtle analysis. It is there to hopefully give the viewer a quick overview of who appears most in the dataset and to anchor the rest of the dashboard. (Few, 2012, 2016) guidance is useful here, for nominal comparison and ranking, bars are hard to beat because the comparison is made through aligned position and length rather than angle or area.

For age profile and housing tenure I used 100% stacked horizontal bars. I did think about grouped bars, but with several categories they became much wider and more cluttered. What I really wanted to show was not absolute size but internal composition. In other words, once a citizenship group is selected, what share falls into each age band, and what share is in each housing tenure category? A proportional stacked bar works reasonably well for that because every row has the same total length, so the viewer can focus on the balance within a group rather than being distracted by raw totals (Financial Times, no date). Using the same visual form for both panels was also intentional. Repeating the idiom makes the dashboard easier to learn because once the viewer understands one compositional chart, the other reads in the same way.

That said, this is also one of the weaker parts of the design. Stacked bars are good for giving an overview, but they are not ideal when someone wants to compare interior segments precisely across many rows. Only the first segment has a shared baseline. The rest are harder to line up visually (Cleveland and McGill, 1984; Few, 2012). So this was a tradeoff, I chose compactness and readability over precision. A small multiples layout or grouped bars would have supported finer comparison, but they would also have taken up more space and made the full dashboard heavier.

For the rent panel I used a point plot rather than another bar chart. I wanted that panel to feel lighter and more direct. A point plot is enough when the task is simply to compare one value per category, and the dashed reference line makes it easy to judge whether a group sits above or below the national average. I preferred this to a bar chart because bars would have added more ink without adding more information (Tufte, 2001). The point plot also gives the dashboard a bit more visual variety without becoming inconsistent. That matters because all four panels need to work together rather than compete with each other. The general principle behind that choice is simple enough, keep the data visually prominent and avoid support structures that do more work than the data itself (Tufte, 2001; Few, 2012; Matthews, 2026b).

In terms of interaction, I kept things intentionally modest. The dropdown filter applies across every panel, so the viewer can choose a citizenship group and then read the rest of the dashboard as a coordinated set. I considered click selection directly on the bars, but it felt less transparent. A dropdown is not exciting, but it is clear. One of the strengths of a dashboard like this is being able to compare views side by side without having to remember what was on a previous screen. The viewer can move their eyes between panels rather than rely on memory, which is one of the reasons dashboards are useful in the first place (Card, Mackinlay and Shneiderman, 1999; Munzner, 2014; Matthews, 2026d, 2026a).

I also ruled out a few alternatives. A pie chart would have been a poor choice for age or tenure because the comparisons would rely on angle and area, which are harder to judge accurately (Cleveland and McGill, 1984; Few, 2012). A choropleth map also did not make much sense, even though migration might initially suggest mapping, because the main variable here is citizenship category rather than geographic position. A map would have looked relevant but not actually helped answer the question. In that sense, one of the more important design decisions was resisting visuals that seemed thematic but did not fit the task (Munzner, 2014).

The dashboard does produce useful insight, even if the findings are not especially dramatic. It becomes obvious quite quickly that citizenship groups do not just differ in number. They also differ in age structure, in the balance between renting and other tenure types, and in how their average rents compare with the wider benchmark. That is really the main value of the dashboard, it brings those differences into one place so they can be read together rather than as disconnected tables. The main limitation is that it still simplifies a lot. Rent is shown as an average, so variation within groups is hidden. The citizenship categories are also only one way of slicing migration, and they do not capture everything someone might care about. The dashboard is therefore better at comparison than explanation. It can show that groups differ but not fully explain why.

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